IDX G9 CHEMISTRY S STUDY GUIDE ISSUE S2 FINALS

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This guide is a combined guide. Due to the final range including content from before the monthlies, we have included content from the previous issue in this guide for your convenience.

8.2 Molecular compounds

I. Molecules and Molecular Compounds

➢ A <u>molecule</u> is a neutral group of atoms joined together by covalent bonds

➢ An oxygen molecule is an example of a diatomic molecule—a molecule that contains two atoms

➢ Molecules can also be made of atoms of different elements

➢ A compound composed of molecules is called a molecular compound

II. Representing Molecules

➢ A **<u>molecular formula</u>** is the chemical formula of a molecular compound

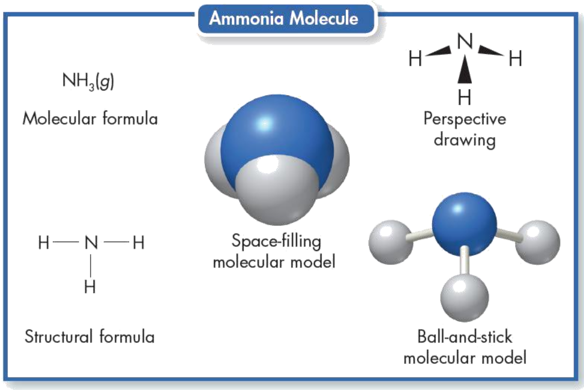
➢ A molecular formula shows how many atoms of each element a substance contains

➢ A molecular formula reflects the actual number of atoms in each molecule

➢ The subscripts are not necessarily the lowest whole-number ratios

➢ Butane (C4H10) is also a molecular compound

➢ molecular formula does not show the molecule’s structure



- Ball-and-stick model shows three-dimensional structure of molecule
- A *ball* represents an atom's nucleus and inner-level electrons
- Sticks represents bonds, Straight stick represents single bonds; curved springs represent multiple bonds
- The arrangement of atoms within a molecule is called its molecular structure

III. Comparing Molecular and Ionic Compounds

- Representative unit:
  - ✓ Molecular compound: molecule
  - ✓ Ionic compound: formula unit
  - ✓ A formula unit is the lowest whole-number ratio of ions in an ionic compound
- Melting point:
  - ✓ Molecular compounds tend to have relatively lower melting and boiling points than ionic compounds
  - ✓ Many molecular compounds are gases or liquids at room temperature
- Types of component element
  - ✓
    - *Metal + Nonmetal → Ionic compound*
    - *Nonmetal + Nonmetal → Molecular compound*

8.3 Bonding Theories

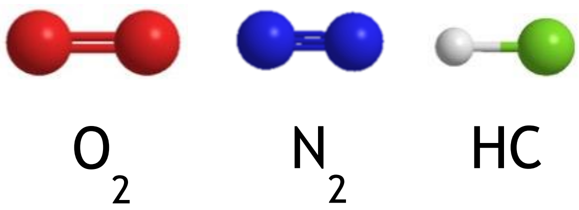
I. VSEPR Theory

➢ In order to explain the three-dimensional shape of molecules, scientists use valence-shell electron-pair repulsion theory (VSEPR theory)

➢ VSEPR theory states that the repulsion between electron pairs causes molecular shapes to adjust so that the valence-electron pairs stay as far apart as possible.

➢ In short, valence electron pairs stay as far apart as possible because they repel.

➢ Linear: In a linear molecule, the atoms can be connected by a straight line

➢ 

➢ Bond angle is the geometric angle between two adjacent bonds

➢ Trigonal Planar: A trigonal planar molecule has a triangular, flat shape (120 degrees angle)

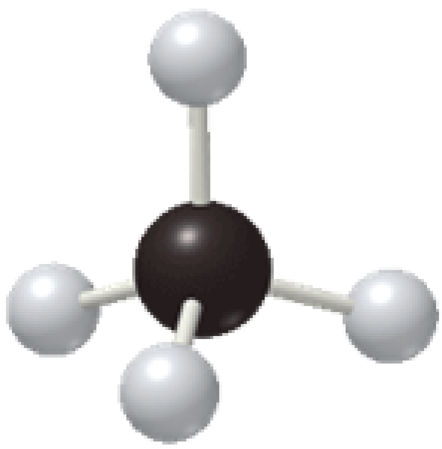
➢ eg: boron trichloride (BCl_3)

➢ The three pairs electrons are arranged as far apart as possible

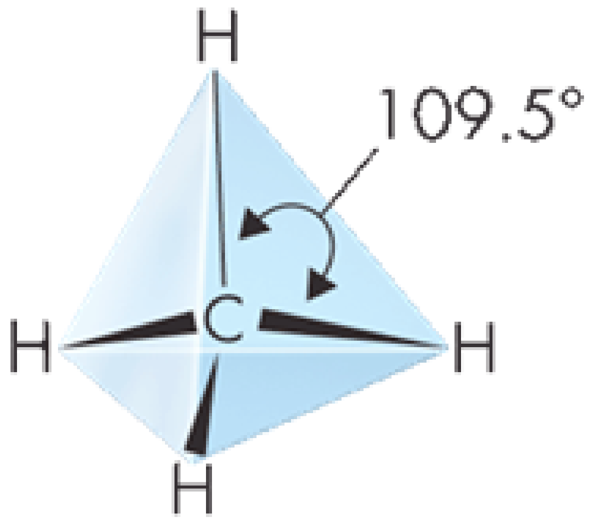
➢ Tetrahedral: A tetrahedral molecule has four surfaces (109.5 degrees angle)

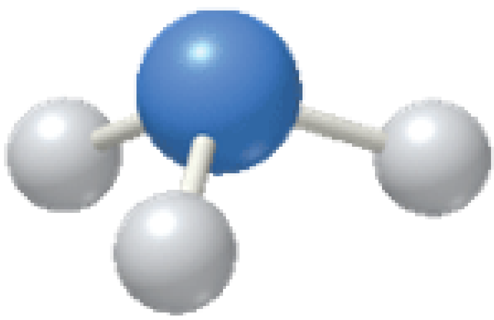
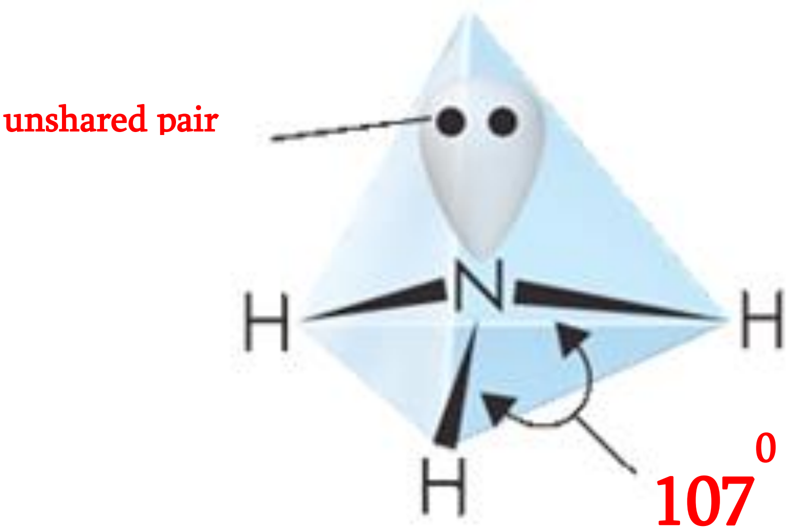
➢ The four pairs electrons are arranged as far apart as possible

➢ The hydrogens in a methane molecule are at the four corners of a geometric solid called a regular tetrahedron

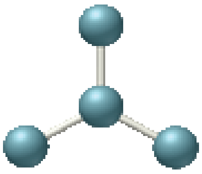
➢ 

➢ In this arrangement, all of the H–C–H angles are 109.5°. This also goes with other tetrahedral molecules.

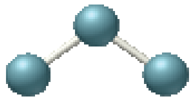
- 

- Pyramidal: The four pairs electrons are arranged as far apart as possible (107 degrees angle)
- The nitrogen in ammonia (NH_3) is surrounded by four pairs of valence electrons, one of which is an unshared pair.
- 
- The unshared pair strongly repels the bonding pairs, pushing them together.
- 

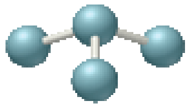
Linear



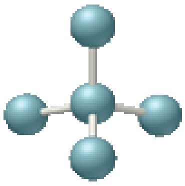
Trigonal planar



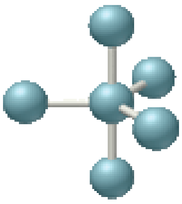
Bent



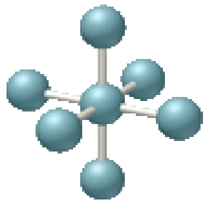
Pyramidal



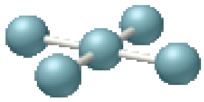
Tetrahedral



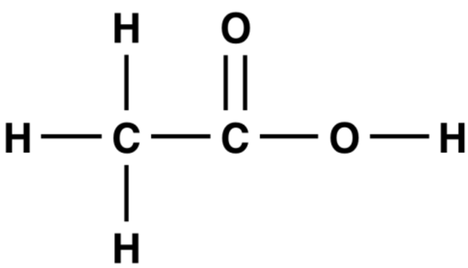
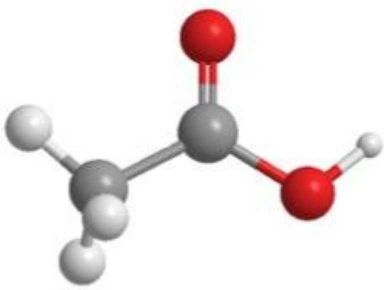
Trigonal bipyramidal



Octahedral



Square planar

- ➢ Both tetrahedral and pyramidal shapes have four pairs of electrons surrounding the central atom, and the unshared pairs exert a greater repulsion force, thus the bond angle in pyramidal is slightly less than in tetrahedral shape.
- ➢ In short words, unshared pairs "takes up more space" and causes the others to "squeeze"
- ➢ Bent: The four pairs electrons are arranged as far apart as possible(two other atoms and has two unshared electrons)
- ➢ The bond angle in bend shape is 105^0
- ➢ Shape of Larger Molecules:
  - ✓ The shape of larger molecules is determined by the bond angle of each central atom.
  - ✓  
  - ✓ eg: bond angle on
  - ✓ 1^{st} carbon is 109.5^0 (tetrahedral)
  - ✓ 2^{nd} carbon is 120^0 (pyramidal)

- ✓ 3^{rd} oxygen is 105^{0} (bent)

II. Hybrid orbitals

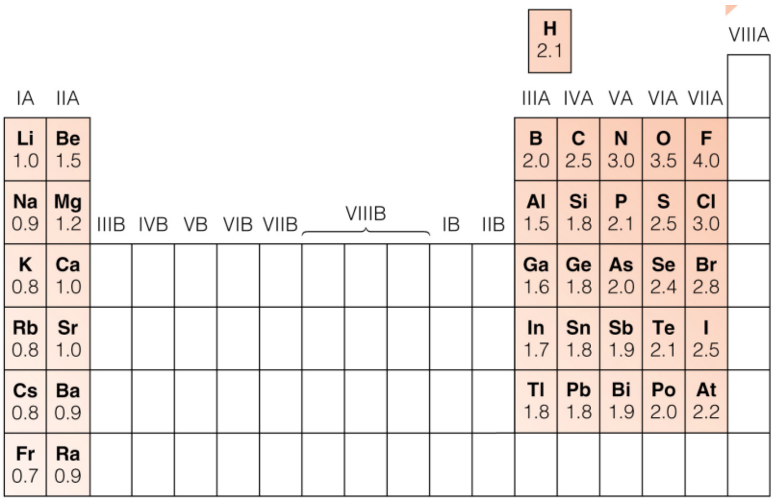
- When an atom approaches another atom to form bond, the atomic orbitals mix together, forming hybrid orbitals
- In hybridization, several atomic orbitals mix to form the same total number of equivalent hybrid orbitals.

Molecular shape	Number of Orbitals	Hybrid Orbitals
Linear	2	sp
Trigonal Planar	3	sp^2
Tetrahedral, pyramidal, bent(105°)	4	sp^3

8.4 Polar bonds and molecules

I. Bond Polarity

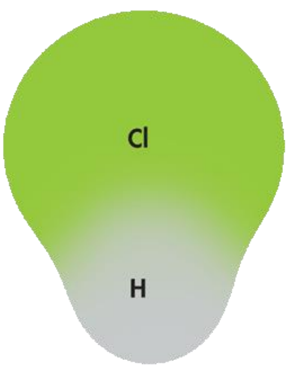
- The bonding pairs of electrons in covalent bonds are pulled between the nuclei of the atoms sharing the electrons
- The nuclei of atoms pull on shared electrons
- A nonpolar bond is a covalent bond formed when the atoms in the bond pull equally and the bond electrons are shared equally
- A polar bond is a covalent bond between atoms in which the electrons are shared equally

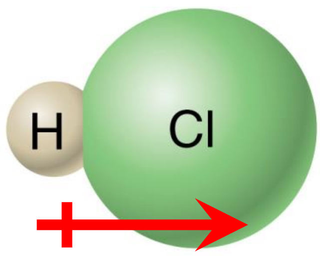
➢ 

➢ Electronegativity Difference	➢ Bond Type
≤0.4	Nonpolar covalent
0.4 < … < 2.0	Polar covalent
≥2.0	ionic

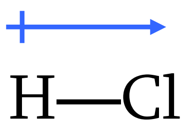
II. Describing polar covalent bonds

- ➢ The lowercase Greek letter delta (δ) denotes that atoms in the covalent bond acquire only partial charges, less than 1+ or 1-
- ➢ These partial charges are shown as clouds of electron density
- ➢ Ex. The chlorine atom attracts the electron cloud more than the hydrogen atom does

➢ 



(a) HCl is polar



- The polar nature of the bond may also be represented by an arrow pointing to the more electronegative atom (the right picture above)

III. Determine the polarity of molecules

- A molecule containing only nonpolar bonds is called a nonpolar molecule
- A molecule containing polar bonds is NOT necessarily a polar molecule
- The **shape of a molecule** and the **polarity of its bonds** together determine whether the molecule is polar or nonpolar

Molecular shape and polarity

Molecular Shape		**Are all "outside" atoms same?**	**Molecular Polarity**
AX_2	linear	**yes**	**Nonpolar**
		no	Polar
	bent	-	polar
AX_3	Trigonal planar	**yes**	**nonpolar**
		no	polar
	pyramidal	-	polar
AX_4	tetrahedral	**yes**	**nonpolar**
		no	polar

IV. Attractions Between Molecules

- Van der Waals Forces: The two weakest attractions between molecules
  - ✓ Also known as Intermolecular force
- **Dipole interactions** occur when polar molecules are attracted to one another

- ✓ The slightly negative region of a polar molecule is weakly attracted to the slightly positive region of another polar molecule
- ✓ similar to, but much weaker than, ionic bonds

➢ **Dispersion forces**, are caused by the motion of electrons

- ✓ Dispersion forces are the weakest of all molecular interactions
- ✓ Dispersion forces operate between all molecules. They occur even between nonpolar molecules
- ✓ The strength of dispersion forces generally increases as the number of electrons in a molecule increases

➢ **Hydrogen bonding**: especially strong dipole-dipole force between the hydrogen atom of one molecule(δ+) and the F, O, S, or N atom of another molecule (δ-).

- ✓ The positive region of one water molecule attracts the negative region of another water molecule
- ✓ Hydrogen bonds are the strongest of the intermolecular forces
- ✓ They are extremely important in determining the properties of water and biological molecules
- ✓ A snowflake's shape is determined by the interactions of hydrogen bonds during its formation
- ✓ Dispersion force → Dipole interaction → Hydrogen bond

  Relative strength of the intermolecular forces

V. Intermolecular Attractions and Molecular Properties

➢ The diversity of physical properties among covalent compounds is mainly because of widely varying intermolecular attractions

➢ The melting and boiling points of most molecular compounds are low compared with those of ionic compounds

➢ In most solids formed by molecules, only the weak attractions between molecules need to be broken.

- A few solids that consist of molecules do not melt until the temperature reaches 1000°C or higher
  - ✓ Most of these very stable substances are **network solids** (or network crystals), solids in which all of the atoms are covalently bonded to each other
- Ex. Diamond
  - ✓ Each carbon atom in a diamond is covalently bonded to four other carbons, interconnecting carbon atoms throughout the diamond
  - ✓ Diamond does not melt; rather, it vaporizes to a gas at 3500° C and above
- Silicon dioxide (quartz) has a giant covalent structure similar to diamond
  - ✓ Each silicon atom (2.8.4) is bonded to four oxygen atoms, and each oxygen atom (2.6) is bonded to two silicon atoms

Characteristics of Ionic and Molecular Compounds		
Characteristic	**Ionic Compound**	**Molecular Compound**
Representative unit	Formula unit	Molecule
Bond formation	Transfer of one or more electrons between atoms	Sharing of electron parts between atoms
Type of elements	Metallic and nonmetallic	Nonmetallic
Physical state	Solid	Solid, liquid, or gas
Melting point	High (usually above 300° C)	Low (usually below 300° C)
Electrical conductivity of aqueous solution	Good conductor(when melted or in solution)	Poor to nonconducting

9.1-9.3 Naming Ionic Compounds and Molecular Compounds

1. Monatomic ions

- consist of a single atom with a positive or negative charge resulting from the loss or gain of one or more valence electrons

- The names of cations are the same as the elements, followed by the word ion or cation.
- Anion names start with the stem of the element name and end in –ide
- Two methods are used to name metals that form more than one ion
  - ✓ Stock system: Place a Roman numeral in parentheses
  - ✓ Ex: Fe^{2+} -> iron(II) ion Fe^{3+} -> iron(III) ion
  - ✓ Root word
  - ✓ Ex: Fe^{2+} -> ferrous ion Fe^{3+} -> ferric ion

2. Polyatomic ions

- Composed of more than one atom
- Behaves as a unit and carry a charge
- ended with suffix –ate for ions with more oxygen atoms. (Example: Chlorate, ClO_3^-)
- ended with suffix –ite for ions with less oxygen atoms (Example: Chlorite, ClO_2^-)
- add prefix –per and plus suffix –ate for ions with greatest number of oxygen atoms (Example: Perchlorate, ClO_4^-)
- add prefix –hypo and plus suffix –ite for ions with fewest number of oxygen atoms (Example: Hypochlorite, ClO^-. Note that -hypo is pronounces the same as "hypa")

3. Binary Ionic Compounds

- To write the formula of a binary ionic compound, first write the symbol of the cation and then the anion. Then add subscripts as needed to balance the charge
- Crisscross method (A_xB_y)
- To name any binary ionic compound, place the cation name first, followed by the anion name
- If the metallic element in a binary ionic compound has more than one common ionic charge, a Roman numeral must be included. Example: Cu^{2+} is Copper (II)

4. Compounds with polyatomic ions

- Whenever more than one polyatomic ion is needed to balance the charges in an ionic compound, use parentheses to set off the polyatomic ion in the formula

5. Binary Molecular Compounds

- Write the names of the elements in the order listed in the formula
- Use prefixes appropriately to indicate the number of each kind of atom

PREFIX	NUMBER
mono-	1
di-	2
tri-	3
tetra-	4
penta-	5
hexa-	6
hepta-	7
octa-	8
nona-	9
deca-	10

- End the name of the second element with the suffix –ide (Example: Carbon Tetrabromide is CBr4)
- If just one atom of the first element is in the formula, omit the prefix mono- for that element (Example: Carbon Monoxide is CO)
- Also, the vowel at the end of a prefix is sometimes dropped when the name of the element begins with a vowel (Example: pentoxide, not pentaoxide)

9.4 Naming and Writing Formulas for Acids and Bases

- Acid is a compound that contains one or more hydrogen atoms and produces hydrogen ions when dissolved in water

1. Anion: ______ide → acid: hydro___ic acid

- 2. Anion: ______ate → acid: ____ic acid
- 3. Anion: ______ite → acid: ____ous acid
- A base is compound that produces hydroxide ions when dissolved in water (generally is an ionic compound)
- The name of the cation is followed by the name of the anion